Operating Systems
Lecture 13

IO devices and disk

Prof. Mengwei Xu

Goals for Today

- I/O Devices
- Storage Devices

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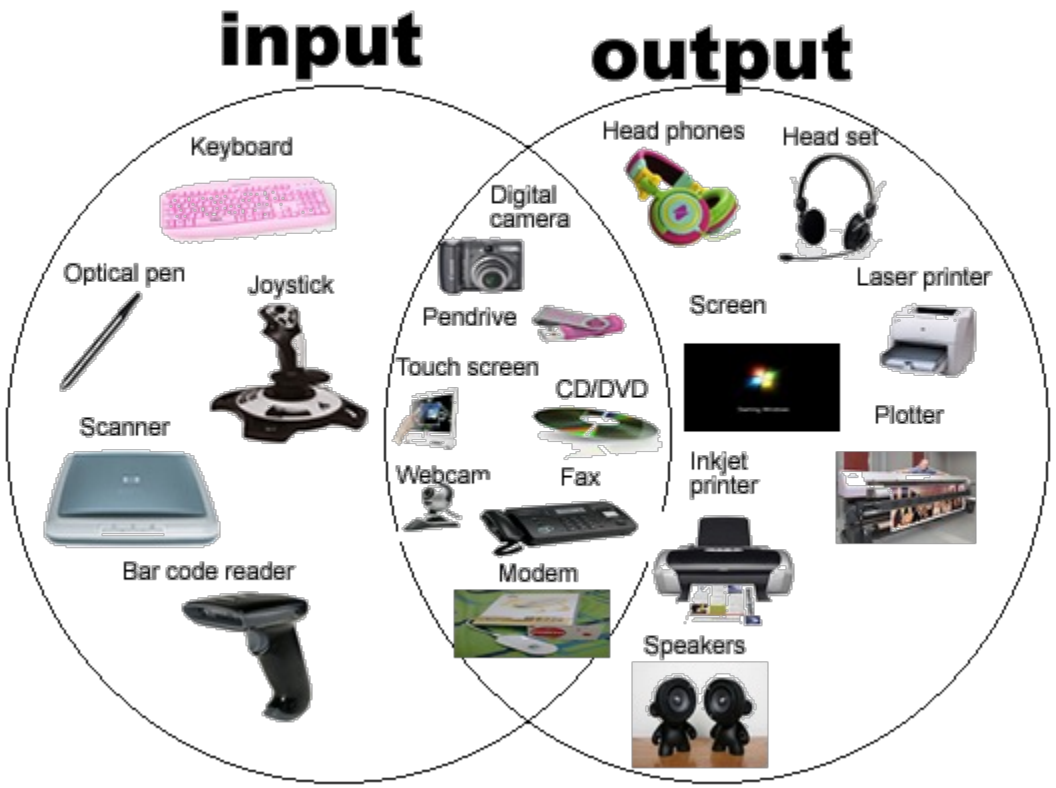
- I/O Devices
- Storage Devices

I/O Devices

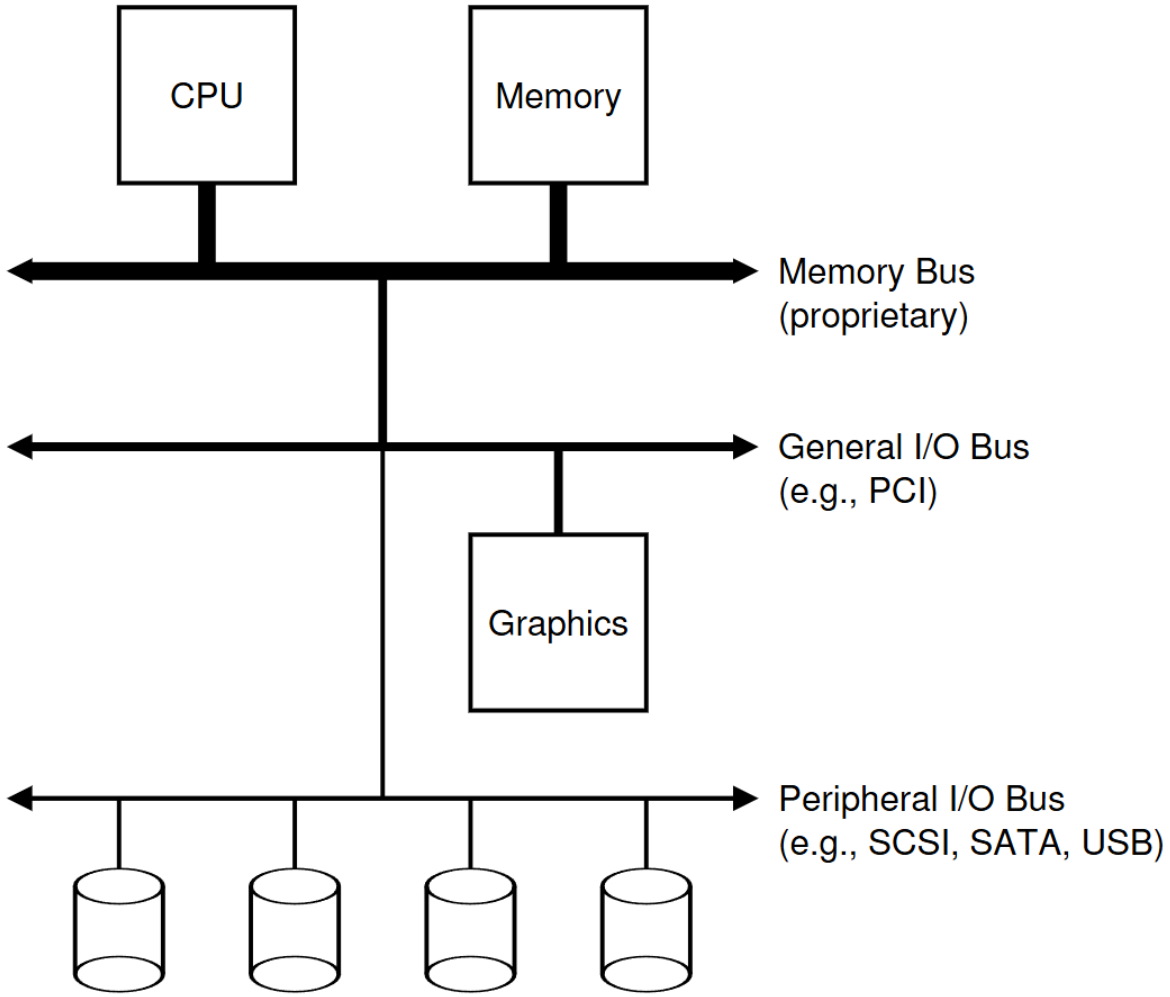
- I/O Devices (输入输出设备) are important to today's computers



- Without input devices, the machine only repeat computations and generate the same output
- Without output devices.. What's the purpose of it?

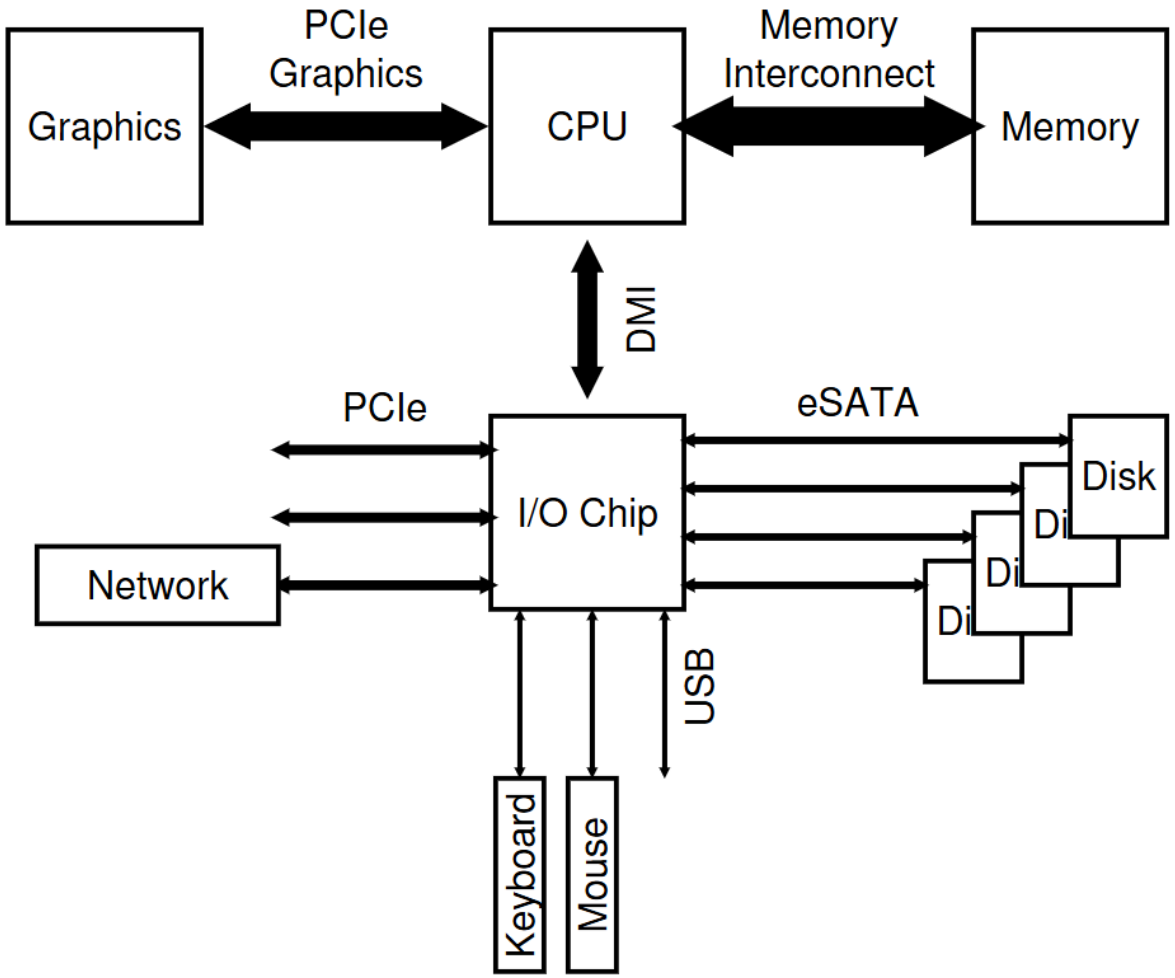
System Architecture

- The old architecture of computer IO



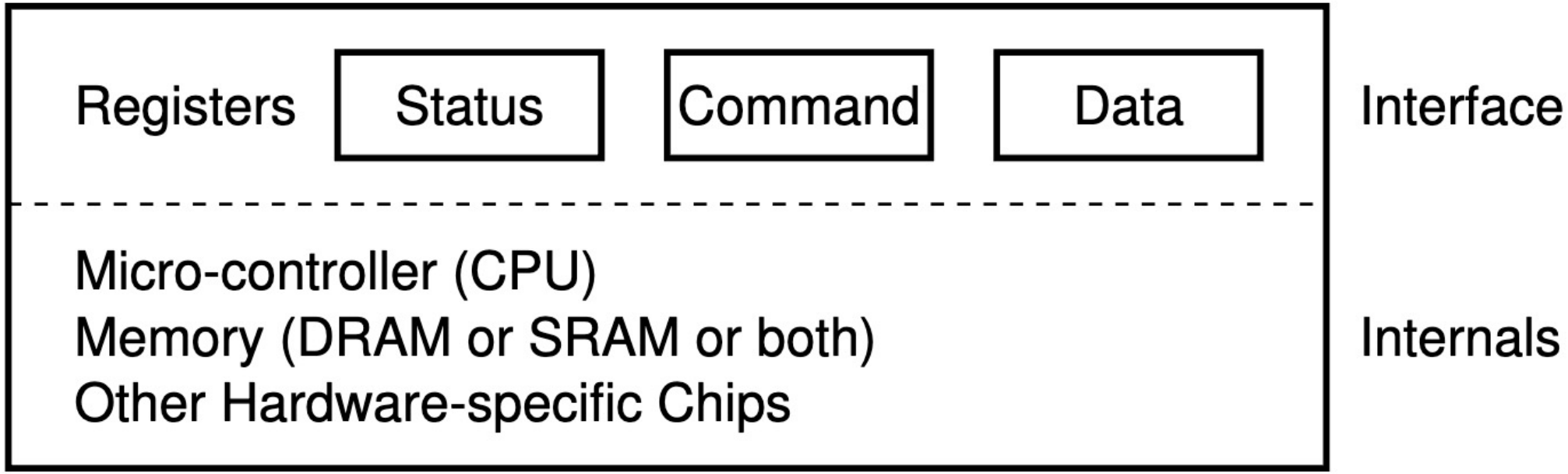
System Architecture

- The modern architecture of computer IO



A Simple IO Device

- Part#1: interface
- Part#2: internal structure
  - Implementation specific and is responsible for implementing the abstraction the device presents to the system.
  - Complex devices could have their own CPU and memory as well.



A Simple IO Device

- a status register, which can be read to see the current status of the device;
- a command register, to tell the device to perform a certain task;
- a data register to pass data to the device, or get data from the device.

```
While (STATUS == BUSY)
    ; // wait until device is not busy
Write data to DATA register
Write command to COMMAND register
    (starts the device and executes the command)
While (STATUS == BUSY)
    ; // wait until device is done with your request
```

A Simple IO Device

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Polling is inefficient!

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A Simple IO Device

- a status register, which can be read to see the current status of the device;
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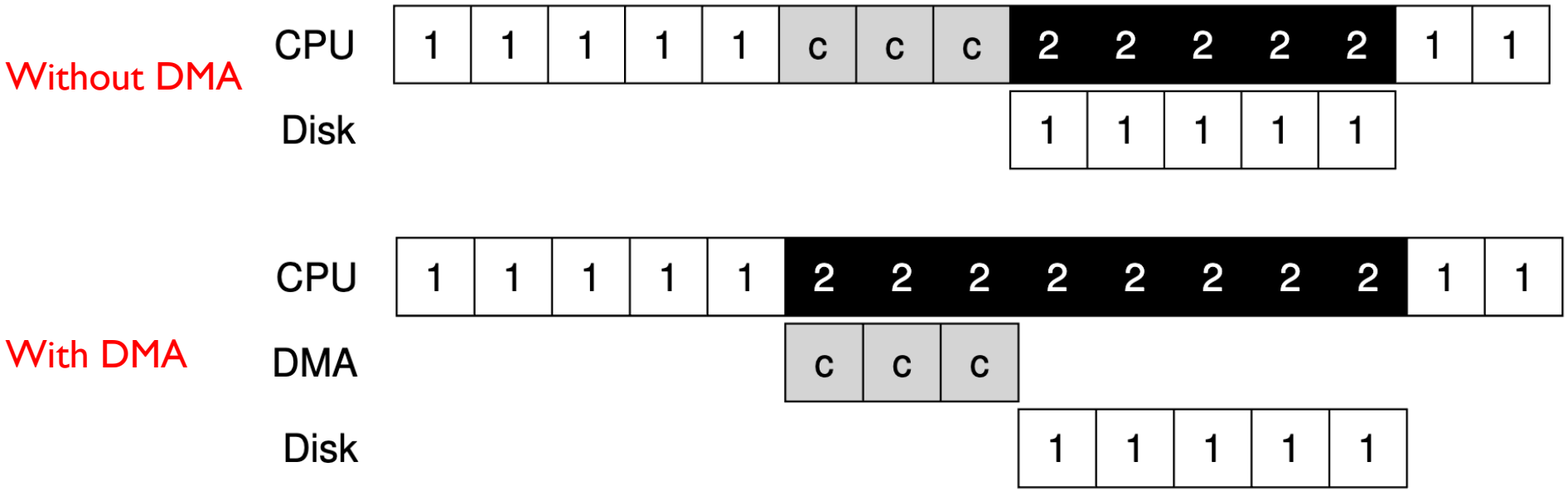
- a data register to pass data to the device, or get data from the device.

- Using interrupts instead of polling

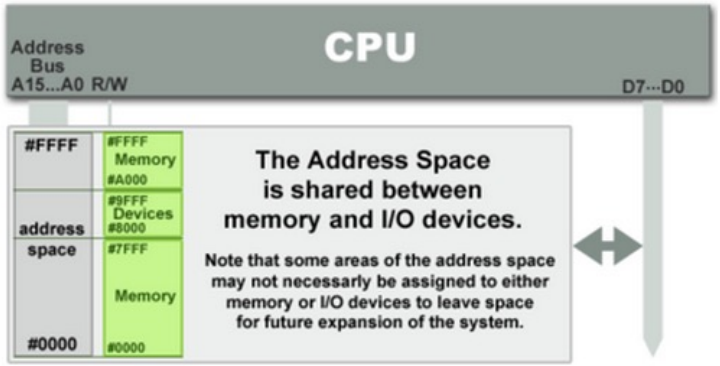
Direct Memory Access

- A DMA engine is a device that can orchestrate transfers between devices and main memory without much CPU intervention.
  - To transfer data to the device, for example, the OS would program the DMA engine by telling it where the data lives in memory, how much data to copy, and which device to send it to. At that point, the OS is done with the transfer and can proceed with other work. When the DMA is complete, the DMA controller raises an interrupt, and the OS thus knows the transfer is complete.

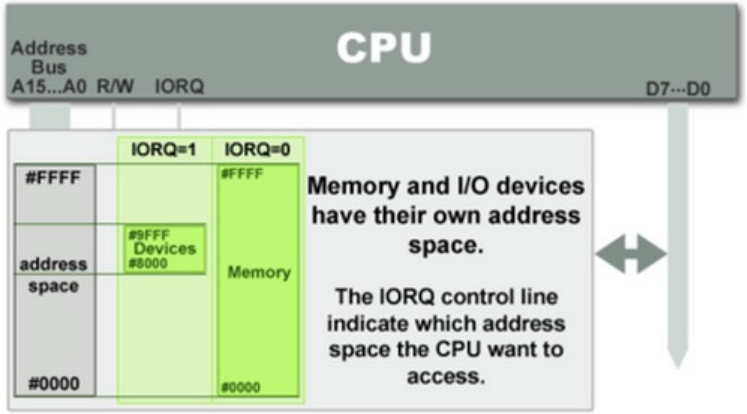


Memory-mapped I/O vs. Port-mapped I/O

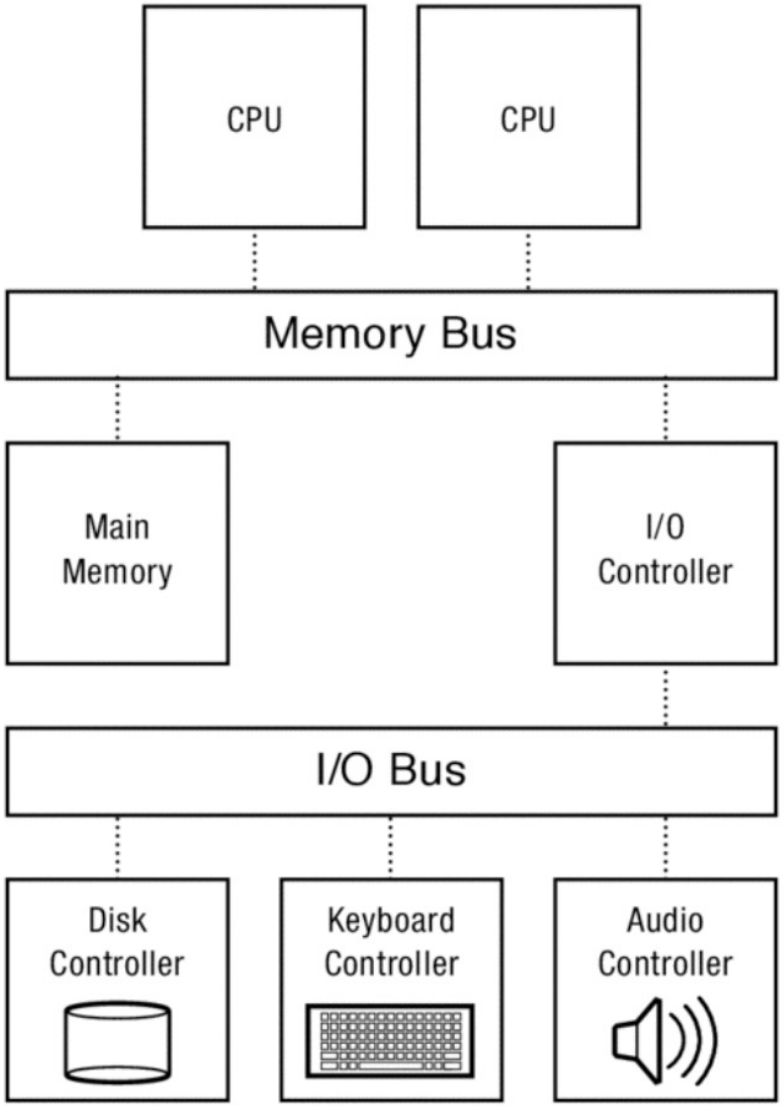
- Two complementary ways for CPU to access I/O devices
  - I/O devices have their own registers (or memory)
- Memory-mapped I/O (MMIO): let memory and devices share the physical address space.
  - Most widely adopted
  - Shared address bus

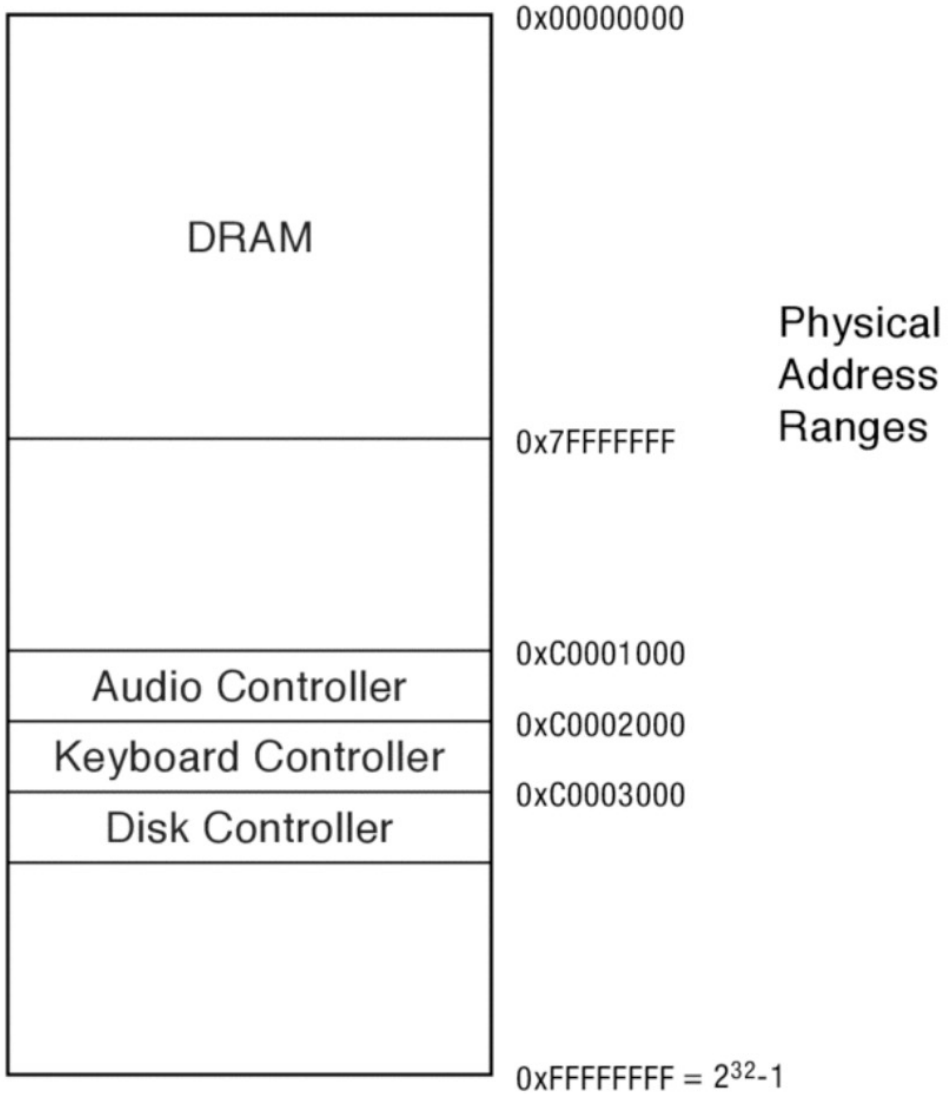


- Port-mapped I/O (PMIO), or isolated I/O: use specialized instructions to R/W I/O devices
  - In Intel: outb, outw, etc.



Storage Stack





A Simple IDE Disk Driver

```
Control Register:
  Address 0x3F6 = 0x08 (0000 1RE0): R=reset,
                  E=0 means "enable interrupt"

Command Block Registers:
  Address 0x1F0 = Data Port
  Address 0x1F1 = Error
  Address 0x1F2 = Sector Count
  Address 0x1F3 = LBA low byte
  Address 0x1F4 = LBA mid byte
  Address 0x1F5 = LBA hi  byte
  Address 0x1F6 = 1B1D TOP4LBA: B=LBA, D=drive
  Address 0x1F7 = Command/status

Status Register (Address 0x1F7):
    7      6     5      4     3     2     1     0
   BUSY  READY FAULT SEEK  DRQ  CORR IDDEX ERROR

Error Register (Address 0x1F1): (check when ERROR==1)
    7      6     5      4     3     2     1     0
   BBK    UNC   MC   IDNF  MCR  ABRT T0NF AMNF

   BBK  = Bad Block
   UNC  = Uncorrectable data error
   MC   = Media Changed
   IDNF = ID mark Not Found
   MCR  = Media Change Requested
   ABRT = Command aborted
   T0NF = Track 0 Not Found
   AMNF = Address Mark Not Found
```

```
static int ide_wait_ready() {
  while (((int r = inb(0x1f7)) & IDE_BSY) || !(r & IDE_DRDY))
    ;  // loop until drive isn't busy
  // return -1 on error, or 0 otherwise
}

static void ide_start_request(struct buf *b) {
  ide_wait_ready();
  outb(0x3f6, 0);                  // generate interrupt
  outb(0x1f2, 1);                  // how many sectors?
  outb(0x1f3, b->sector & 0xff);   // LBA goes here ...
  outb(0x1f4, (b->sector >> 8) & 0xff);   // ... and here
  outb(0x1f5, (b->sector >> 16) & 0xff);  // ... and here!
  outb(0x1f6, 0xe0 | ((b->dev&1)<<4) | ((b->sector>>24)&0x0f));
  if(b->flags & B_DIRTY){
    outb(0x1f7, IDE_CMD_WRITE);    // this is a WRITE
    outsl(0x1f0, b->data, 512/4);  // transfer data too!
  } else {
    outb(0x1f7, IDE_CMD_READ);     // this is a READ (no data)
  }
}
```

Polling or Interrupt?

Memory-mapped IO or port-mapped IO?

Code from xv6

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  outb(0x1f3, b->sector & 0xff);   // LBA goes here ...
  outb(0x1f4, (b->sector >> 8) & 0xff);   // ... and here
```

```
void ide_rw(struct buf *b) {
  acquire(&ide_lock);
  for (struct buf **pp = &ide_queue; *pp; pp=&(*pp)->qnext)
    ;                              // walk queue
  *pp = b;                         // add request to end
  if (ide_queue == b)              // if q is empty
    ide_start_request(b);          // send req to disk
  while ((b->flags & (B_VALID|B_DIRTY)) != B_VALID)
    sleep(b, &ide_lock);           // wait for completion
  release(&ide_lock);
}

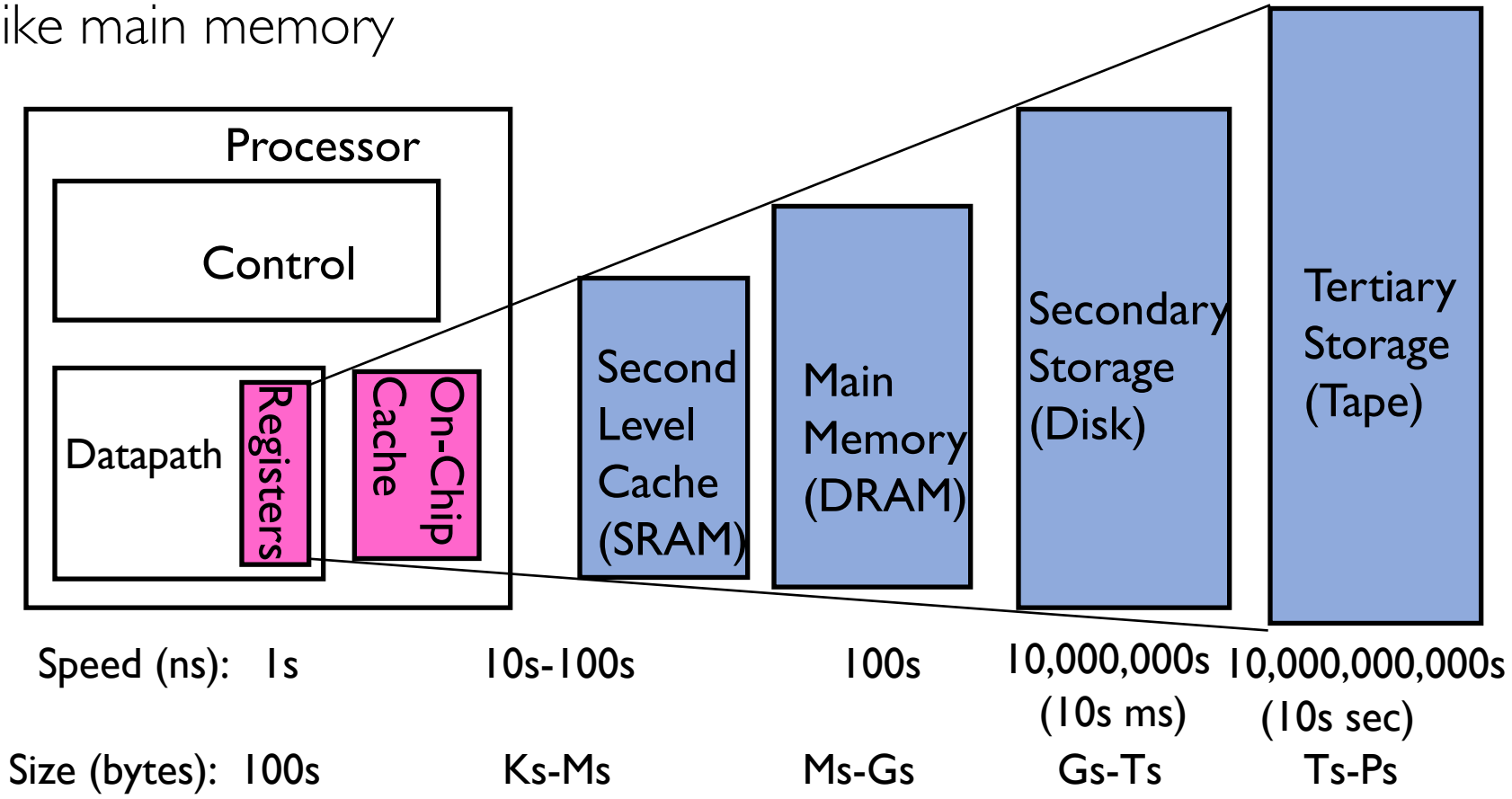
void ide_intr() {
  struct buf *b;
  acquire(&ide_lock);
  if (!(b->flags & B_DIRTY) && ide_wait_ready() >= 0)
    insl(0x1f0, b->data, 512/4);   // if READ: get data
  b->flags |= B_VALID;
  b->flags &= ~B_DIRTY;
  wakeup(b);                       // wake waiting process
  if ((ide_queue = b->qnext) != 0) // start next request
    ide_start_request(ide_queue);  // (if one exists)
  release(&ide_lock);
}
```

Goals for Today

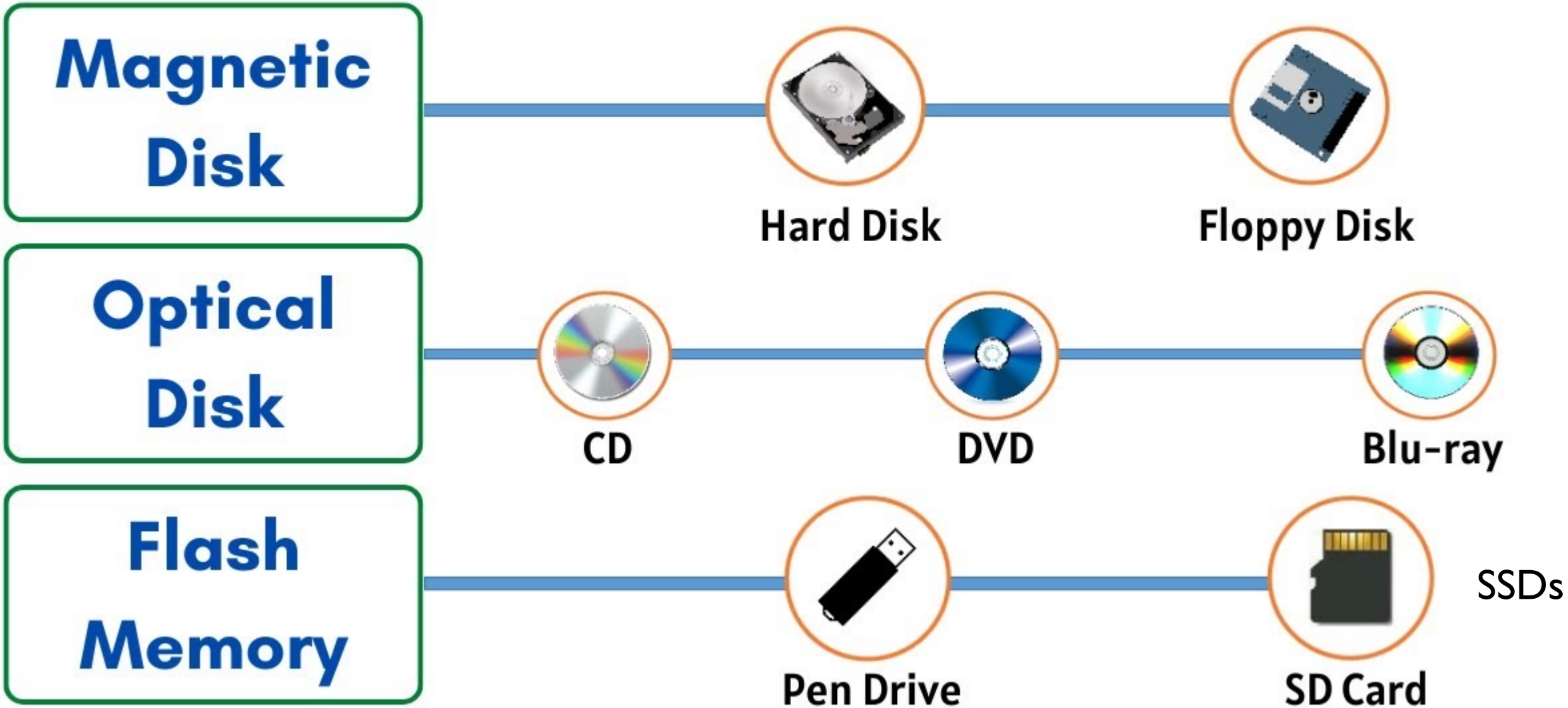
- I/O Devices
- Storage Devices

Storage Devices

- Why we learn the hardware characteristics? Because they help us build better OSes and applications!
- As secondary storage to computers, storage devices are persistent.
  - Unlike main memory



Secondary Storage

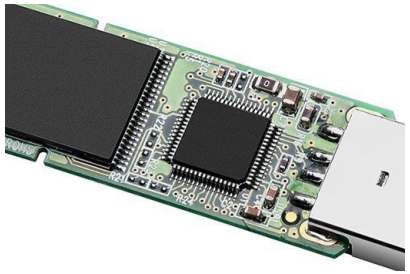


Storage Devices

1. Magnetic disks (磁盘)
   - Storage that rarely becomes corrupted
   - Large capacity at low cost
   - Block level random access
   - Slow performance for random access
   - Better performance for sequential access

Servers, workstations, and labtops

2. Flash memory (闪存)
   - Storage that rarely becomes corrupted
   - Capacity at intermediate cost (5-20x disk)
   - Block level random access
   - Good performance for reads; worse for random writes
   - Erasure requirement in large blocks
   - Wear patterns issue



Smartphones and tablets

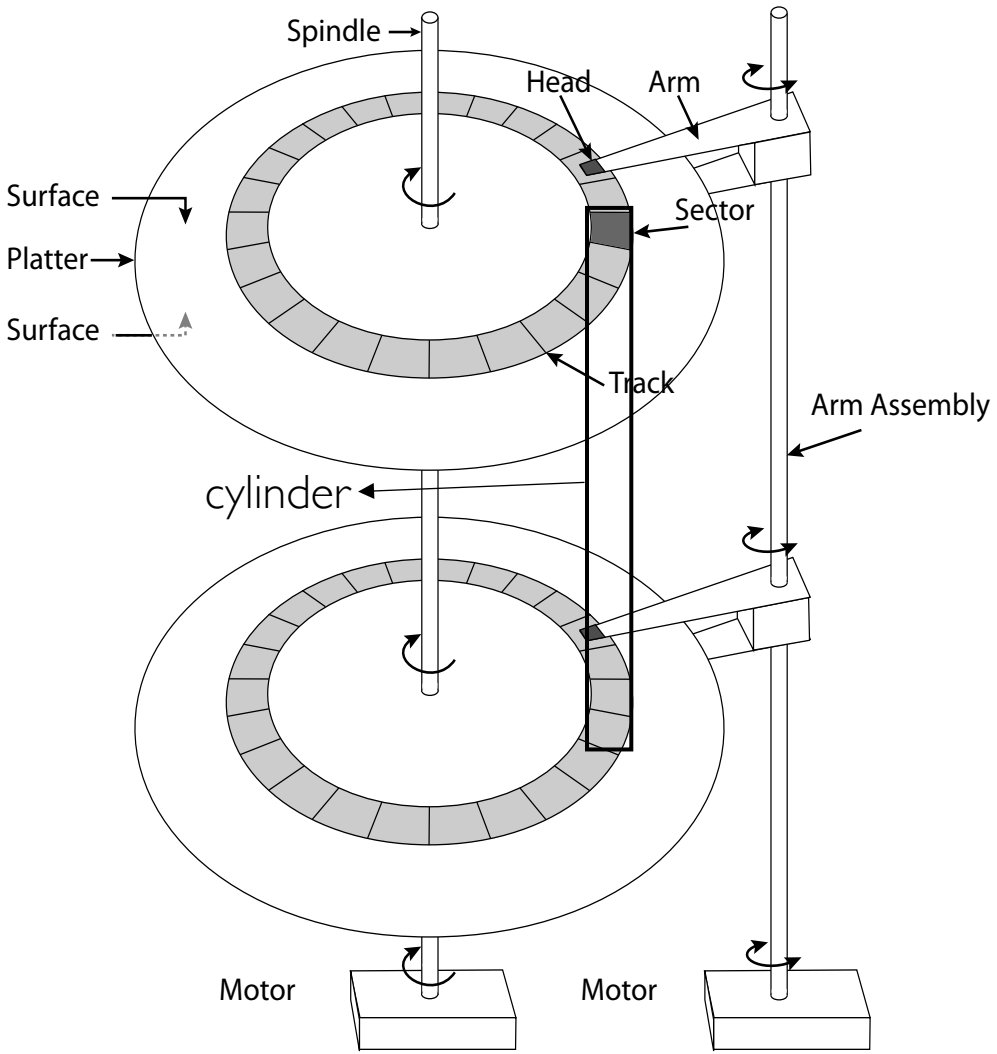
The Magnetic Disk

- Sector (扇区): the unit of transfer
- Track (磁道): ring of sectors
  - ~ 1um (10^{-6}m) wide
    - ❑ Resolution of human eye: 50um
    - ❑ Wavelength of light is ~0.5um
- Cylinder (柱面): stacked tracks
- Head (磁头): attached to movable arms to read data
  - 2 per each platter (磁片) for each surfaces

- Storage capacity =

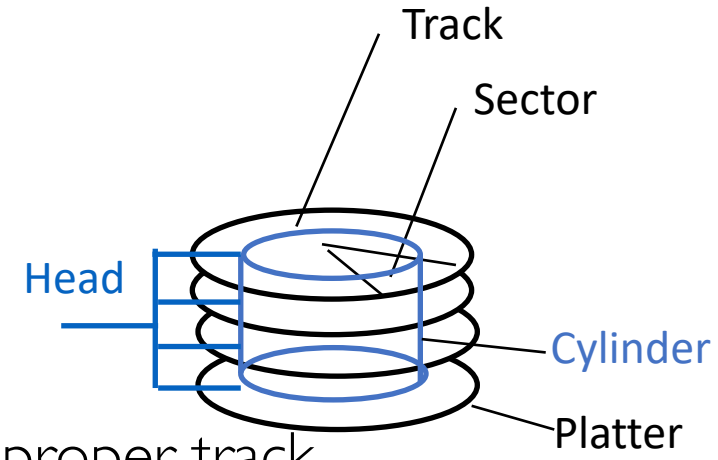
(head #) * (cylinder #) * (sector #) * (sector size)

Often 512 bytes

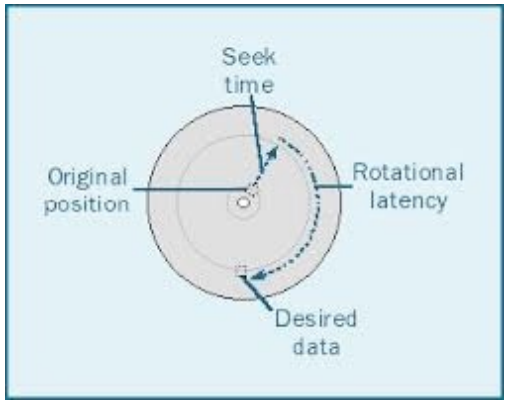


The Magnetic Disk

- Cylinders: all the tracks under the head at a given point on all surface



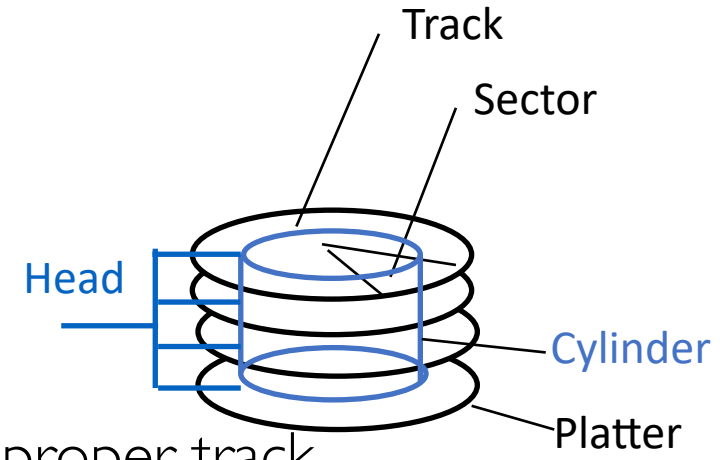
- Read/write data is a three-stage process:
  - Seek time (寻道时间): position the head/arm over the proper track
  - Rotational latency (延迟时间): wait for desired sector to rotate under r/w head
  - Transfer time (传输时间): transfer a block of bits (sector) under r/w head



Seek time = 4-8ms
One rotation = 1-2ms
(3600-7200 RPM)

The Magnetic Disk

- Cylinders: all the tracks under the head at a given point on all surface



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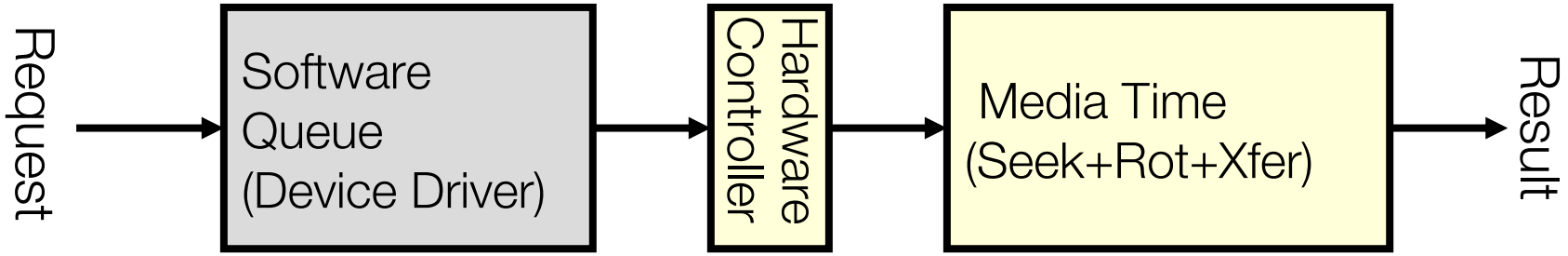
Disk Latency = Queuing Time + Controller time + Seek Time + Rotation Time + Transfer Time



Disk Performance Example

- Assumptions:
  - Ignoring queuing and controller times for now
  - Avg seek time of 5ms,
  - 7200RPM ⇒ Time for rotation: 60000 (ms/minute) / 7200(rev/min) ~= 8ms
  - Transfer rate of 4MByte/s, sector size of 1 Kbyte ⇒
    1024 bytes/4×10^6 (bytes/s) = 256×10^{-6} sec ≅ .26 ms
- Read sector from random place on disk:

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- Read sector from random place on disk:
  - Seek (5ms) + Rot. Delay (4ms) + Transfer (0.26ms) = 9.26ms
  - *Approx* 10ms to fetch/put data: **100 KByte/sec**
- Read sector from random place in same cylinder:

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  - *Approx* 10ms to fetch/put data: **100 KByte/sec**
- Read sector from random place in same cylinder:
  - Rot. Delay (4ms) + Transfer (0.26ms) = 4.26ms
  - *Approx* 5ms to fetch/put data: **200 KByte/sec**
- Read next sector on same track:

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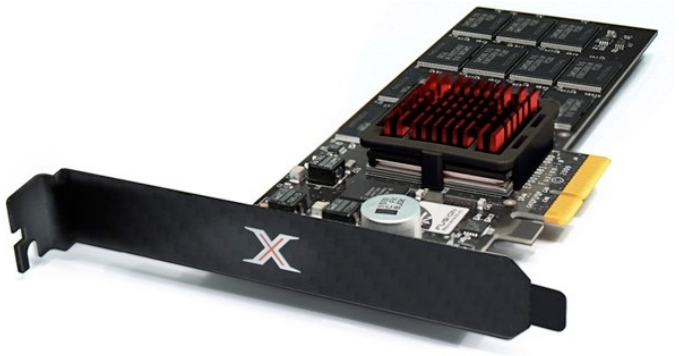
  - Rot. Delay (4ms) + Transfer (0.26ms) = 4.26ms
  - *Approx* 5ms to fetch/put data: **200 KByte/sec**
- Read next sector on same track:
  - Transfer (0.26ms): **4 MByte/sec**

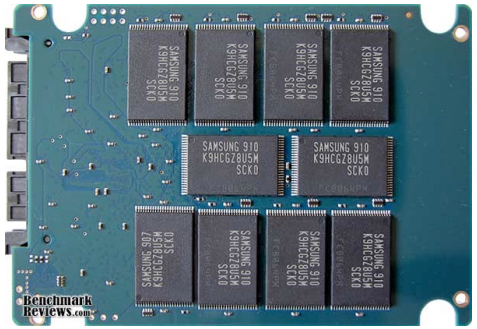
Key to using disk effectively (especially for file systems) is to *minimize seek and rotational delays*

(Lots of) Intelligence in the Controller

- Sectors contain sophisticated error correcting codes
  - Disk head magnet has a field wider than track
  - Hide corruptions due to neighboring track writes
- Sector sparing
  - Remap bad sectors transparently to spare sectors on the same surface
- Slip sparing
  - Remap all sectors (when there is a bad sector) to preserve sequential behavior
- Track skewing
  - Sector numbers offset from one track to the next, to allow for disk head movement for sequential ops
- …

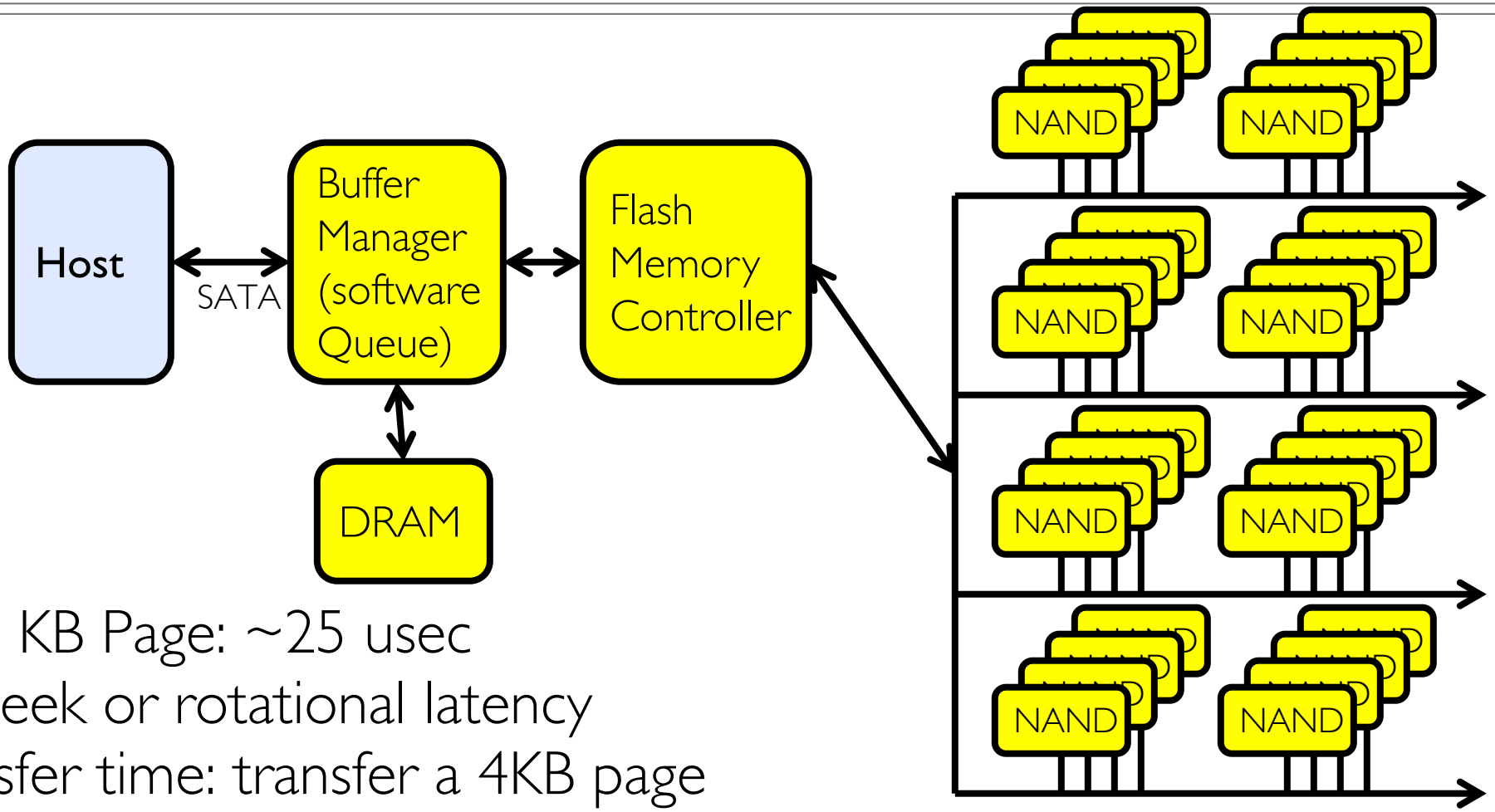
Solid State Disks (SSDs)





- 1995 – Replace magnetic media with non-volatile memory (battery backed DRAM)
- 2009 – Use NAND Multi-Level Cell (2 or 3-bit/cell) flash memory
  - Sector (4 KB page) addressable, but stores 4-64 "pages" per memory block
  - Trapped electrons distinguish between 1 and 0
- No moving parts (no rotate/seek motors)
  - Eliminates seek and rotational delay (0.1-0.2ms access time)
  - Very low power and lightweight
  - Limited "write cycles"
- Rapid advances in capacity and cost ever since!
- A 5-min video on SSD: https://www.bilibili.com/video/BV1644y157mB

SSD Architecture – Reads

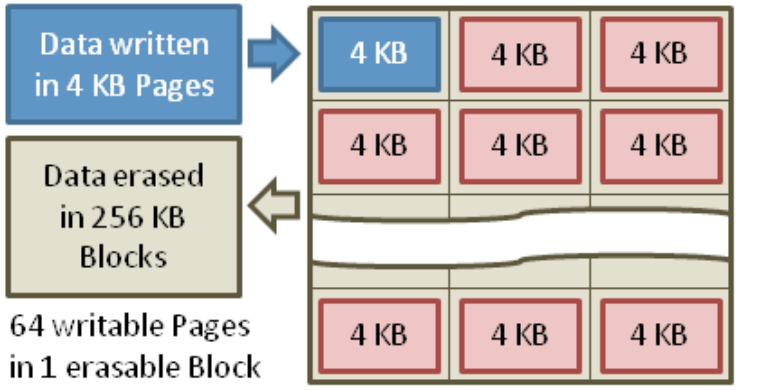


Read 4 KB Page: ~25 usec

- No seek or rotational latency
- Transfer time: transfer a 4KB page
  - ❑ SATA: 300-600MB/s => ~4 x10^3 b / 400 x 10^6 bps => 10 us
- Latency = Queuing Time + Controller Time + Xfer Time
- Highest Bandwidth: Sequential OR Random reads

SSD Architecture – Writes

- Writing data is complex! (~200µs – 1.7ms)
  - Can only write empty pages in a block
  - Erasing a block takes ~1.5ms
  - Controller maintains pool of empty blocks by coalescing used pages (read, erase, write), also reserves some % of capacity
- Rule of thumb: writes 10x reads, erasure 10x writes



Typical NAND Flash Pages and Blocks

https://en.wikipedia.org/wiki/Solid-state_drive

Amusing calculation: is a full Kindle heavier than an empty one?

- Actually, "Yes", but not by much
- Flash works by trapping electrons:
  - So, erased state lower energy than written state
- Assuming that:
  - Kindle has 4GB flash
  - ½ of all bits in full Kindle are in high-energy state
  - High-energy state about 10^{-15} joules higher
  - Then: Full Kindle is 1 attogram (10^{-18}gram) heavier (Using $E = mc^2$)
- Of course, this is less than most sensitive scale can measure (it can measure 10^{-9} grams)
- Of course, this weight difference overwhelmed by battery discharge, weight from getting warm, ….
- According to John Kubiatowicz (New York Times, Oct 24, 2011)

SSD Summary

- Pros (vs. hard disk drives):
  - Low latency, high throughput (eliminate seek/rotational delay)
  - No moving parts:
    - ❑ Very light weight, low power, silent, very shock insensitive
  - Read at memory speeds (limited by controller and I/O bus)
- Cons
  - Small storage (0.1-0.5x disk), expensive (3-20x disk)
    - ❑ Hybrid alternative: combine small SSD with large HDD

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- Cons
  - ~~Small storage (0.1-0.5x disk)~~, expensive (3-20x disk)
    - ❑ Hybrid alternative: combine small SSD with large HDD
  - Asymmetric block write performance: read pg/erase/write pg
    - ❑ Controller garbage collection (GC) algorithms have major effect on performance
  - Limited drive lifetime
    - ❑ 1-10K writes/page for MLC NAND
    - ❑ Avg failure rate is 6 years, life expectancy is 9–11 years
- These are changing rapidly!

No longer true!

Enterprise

10 TB (2016)

- 7 platters, 14 heads
- 7200 RPMs
- 6 Gbps SATA /12Gbps SAS interface
- 220MB/s transfer rate, cache size: 256MB
- Helium filled: reduce friction and power usage
- Price: $500 ($0.05/GB)

IBM Personal Computer/AT (1986)

- 30 MB hard disk
- 30-40ms seek time
- 0.7-1 MB/s (est.)
- Price: $500 ($17K/GB, 340,000x more expensive !!)

Largest SSDs

- 60TB (2016)
- Dual port: 16Gbs
- Seq reads: 1.5GB/s
- Seq writes: 1GB/s
- Random Read Ops (IOPS): 150K
- Price: ~ $20K ($0.33/GB)

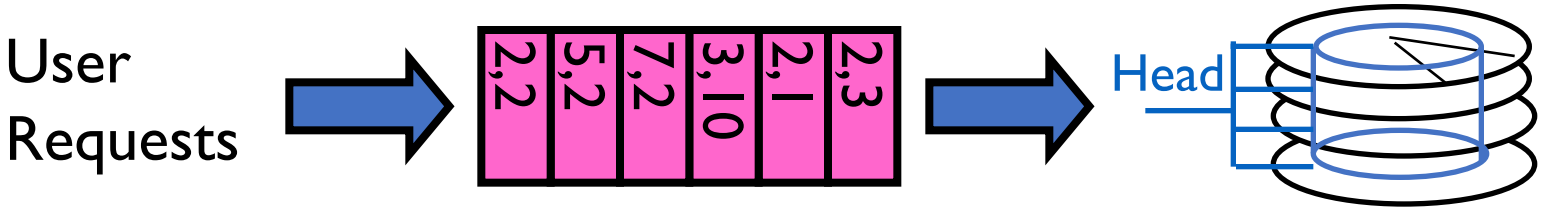
USB Drive

1GB~8GB, 2010

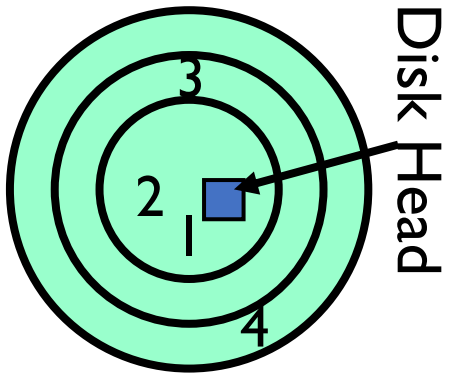
Up to 1TB, 2023

Disk Scheduling

- Disk can do only one request at a time; What order do you choose to do queued requests?
  - The scheduling can be done in OS, firmware, or both.



- FIFO Order
  - Fair among requesters, but order of arrival may be to random spots on the disk ⇒ Very long seeks



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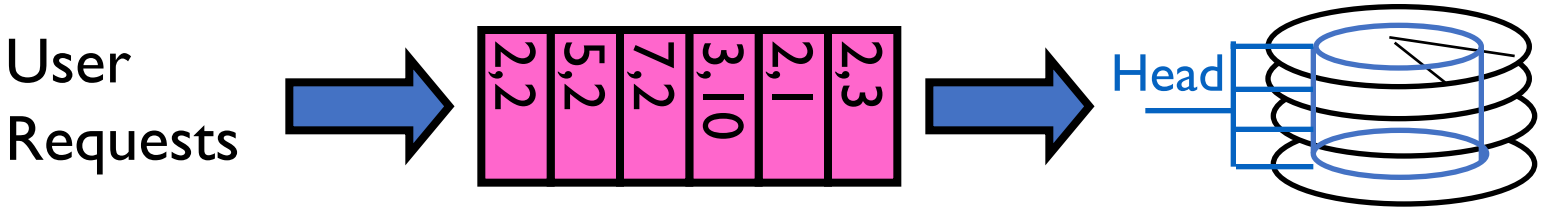
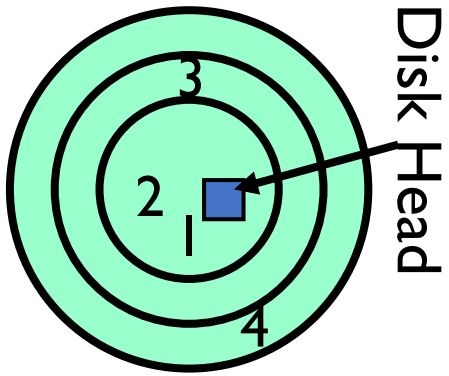


- SSTF: Shortest seek time first
  - Pick the request that's closest on the disk
  - Although called SSTF, today must include rotational delay in calculation, since rotation can be as long as seek
  - Con: SSTF good at reducing seeks, but may lead to starvation



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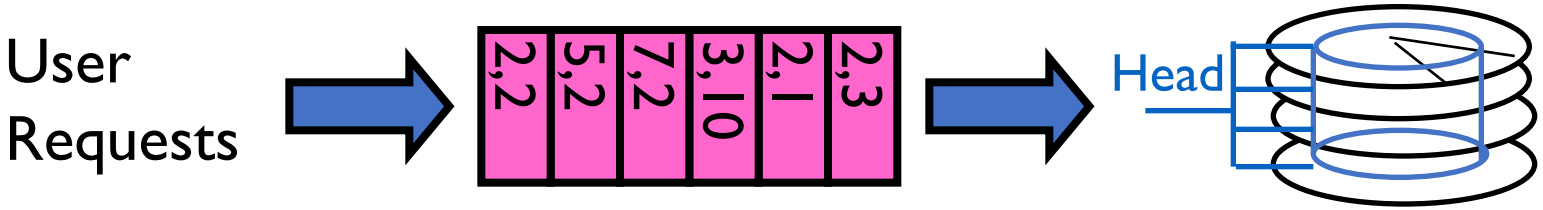
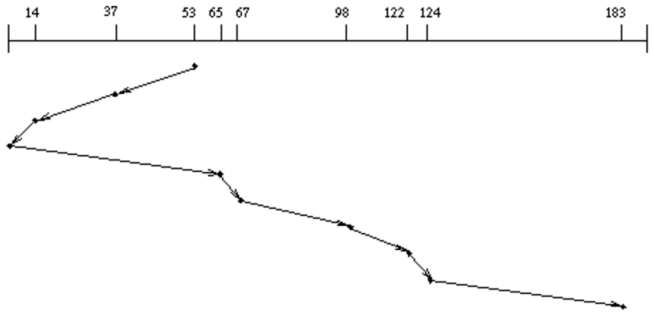
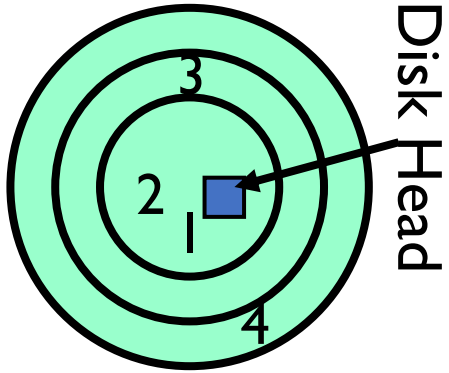


- SCAN: Implements an Elevator Algorithm (电梯算法): take the closest request in a fixed direction of travel (reversed at the end)
  - No starvation, but retains flavor of SSTF





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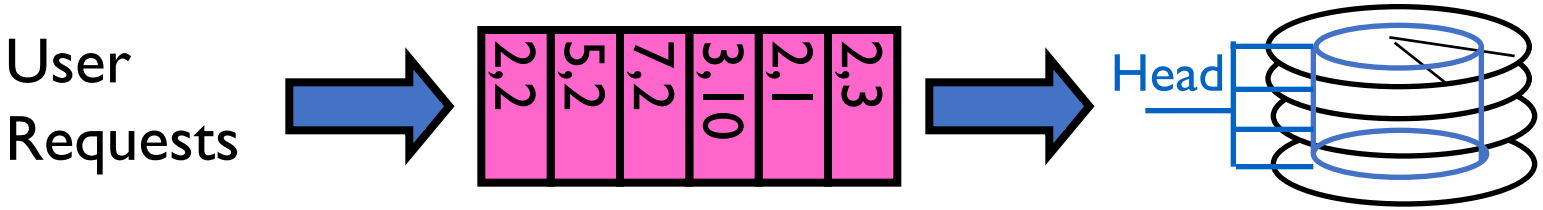
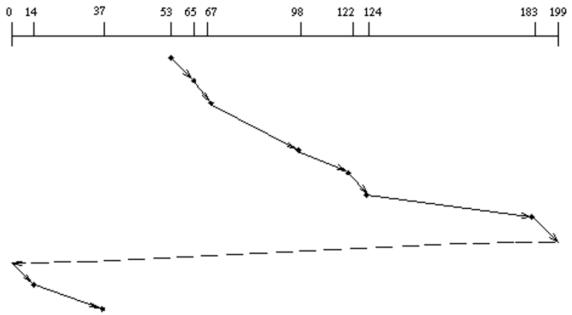
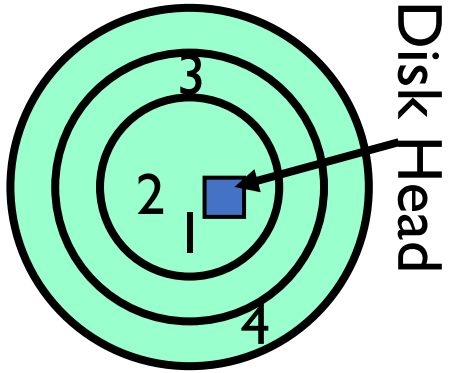


- C-SCAN: Circular-Scan: only goes in one direction
  - Skips any requests on the way back
  - Fairer than SCAN, not biased towards pages in middle





A Simple Read() Lifecycle

- A process issues a syscall read()
- OS moves the calling thread to a wait queue (state=WAITING)
- OS uses memory-mapped I/O to tell the disk to read the requested data and set up DMA so the disk can place the data in kernel's memory
- Disk reads the data and DMAs it into main memory
- Disk triggers an interrupt
- OS's interrupt handler copies the data from the kernel's buffer into the process's address space
- OS moves the thread to the ready list
- The thread is scheduled on CPU, and returns from the read()